June 15, 2025

To whom it may concern,

Recorded Health Inc. is a U.S.-based software company committed to developing platform-based solutions that empower patients, healthcare providers, and payers to consolidate, integrate, analyze, and collaborate around patient-level clinical, operational, and financial data. Our solutions leverage modern data representation architectures and AI-powered analytics to drive better, faster, and smarter decision-making. Data privacy, security, and regulatory compliance are foundational to our work. At our core, we believe that operationalizing data in a patient-centered, secure, and scalable manner is essential for meaningful change in healthcare delivery.

The U.S. healthcare system is in urgent need of a shift toward a model that prioritizes patient sovereignty, data control, and transparent data representation—ultimately flipping the existing power dynamic. We are honored to work toward these goals and to help catalyze a reimagined healthcare future. Recorded Health, along with our solution partners, is pleased to provide our perspectives to inform CMS and ASTP/ONC in their efforts to assess, adopt, and deploy patient-empowerment technologies. We view this initiative as mission critical—a necessary catalyst for a true U.S. healthcare reformation.

Our response to this RFI is based on experience gathered over decades of hands-on patient care, the development of technologies that measurably improve clinical, operational, and financial outcomes, and translational research that bridges innovation and real-world application. We hope our insights and recommendations contribute meaningfully to CMS and ASTP/ONC's mission to deliver ever-better, ever-improving health and wellbeing to the individuals and communities they serve.

Visit Recorded Health Inc, at www.recordedhealth.com for more information.

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

- PC-1 Response:
  - A unified digital, mobile, platform-based ecosystem with functionalities facilitating individual, patient-level health and health-related data acquisition, consolidation, and normalization. This platform would integrate and implement advanced AI data management capabilities and would be consistent with current national trends advocating for and supporting patient-directed personal health data transfer, access, and ownership.

a. What are the top things you would like to be able to do for you or your loved ones' health that can be enabled by digital health products?

- PC-1a Response:
  - Consolidate all health and health-related data within a new AI-enhanced digital, platform-based ecosystem.
  - Allow secure, mobile, on-demand access to all personal health information.
  - Directly query (ask questions) the health records using natural language aided by dedicated fine-tuned and prompt augmented LLMs positioned securely within the "fire wall" of the new digital app platform.
  - Provide the capability to record, review, remember, better understand, and save the important live conversations with healthcare providers. This would represent an extremely valuable patient education tool. If patients could remember and better understand these important conversations; engagement, compliance, empowerment, and enhanced personal agency should improve significantly. These functionalities provide a foundational basis for greatly improved shared decision making.

- Provide a continuously updated summary of key health issues based on the evolving longitudinal personal health and health-related information.
- Provide the capability to digitally share personal health information at the patient's discretion.

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

- PC-1b Response:
  - Routinely monitor, analyze, and reason over complete health record and report new insights, recommendations, and key relationships.
  - Provide access to all audio, text transcripts and summaries of personal interactions with health providers. This information would be date and time stamped and indexed to specific health issues and specific providers for easy access, review, and to achieve better understanding.
  - Assist with direct querying of the personal medical record using natural language.
  - Provide reminders of important current medications, healthy habits, and key personal health issues.
  - Track the progress of specific personal health journeys related to specific medical conditions indexed to accepted guidelines and real-world best practices.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

- PC-2 Response: No

c. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

- PC-2b Response:
  - All contexts and workflows would benefit from a single platform allowing access to all relevant health information. Universal value would be gained from a single portal or single App Platform from which all personal health information is easily accessed and formatted in a contextually relevant, operational, and actionable data representation architecture.

d. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

- PC-2c Response:
  - Legacy EHR data remains highly fragmented and siloed across various delivery platforms as well as, different EMR systems. Objective data (X-rays, lab, and test results, etc.) while generally available, are not annotated with relevant and meaningful context, including meaningful relationships between the data, often resulting in confusion, misinterpretation, and un-necessary anxiety. Health data must be represented in a contextually relevant, understandable, and operational/actionable format.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

- PC-3 Response: Yes. Two excellent patient-centered, feature-rich, digital, mobile, cloud-native platforms.

PC-4. What features are missing from apps you use or that you are aware of today?

- PC-4 Response:
  - Contextual awareness of and relevant multidimensional relationships within the interrelated data.
  - Structured on-demand health record queries using natural language, advanced LLMs and natural language processing technology (NLP).
  - Advanced, machine-learning deterministic analysis and reasoning over the complete longitudinal health record, facilitating on-demand personal health insights.
  - Advanced data, information and knowledge representation architecture (ontology-governed personal clinical knowledge graphs).

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

- PC-4a Response:
  - A platform-based ecosystem whose central focus is the individual patient, providing on-demand access to the consolidated and evolving personal health record. This novel, digital personal health record platform would represent the complete longitudinal, living, and evolving patient-level health record using an advanced, ontology-governed, personal clinical knowledge graph architecture. This knowledge management system supports patient-level AI machine-learning data analysis, reasoning models and natural language querying by fine-tuned, prompt-augmented, secure LLMs. This advanced digital knowledge management system does not exist because the legacy EHR data representation architecture does not support advanced data integration, logical data interoperability, and advanced machine learning analysis and reasoning.

b. What set of workflows do you believe CMS is uniquely positioned to offer?

- PC-4b Response:
  - Utilizing new and novel health information representation architecture (ontology-governed, personal clinical knowledge graphs), over which advanced AI machine learning algorithms and fine-tuned proprietary LLM's can analyze, infer, and reason, CMS could provide their beneficiaries answers to the universal question ALL PATIENTS have:
    - "What are the best treatment options available for people like me who have the same medical condition(s) that I have?"
    - "What works for who, (why, how, and when) and what are the tradeoffs?"

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

- PC-5a Response:
  - Facilitate thorough and exhaustive discussion and dialogue between "CMS" and App Platform Development Teams focusing on the foundational health information representation architecture, academic research fidelity and veracity, model scalability, model reproducibility, model sustainability, model robustness, cybersecurity infrastructure robustness, privacy compliance

infrastructure, etc. Increased patient and caregiver interest in and adoption of patient-centered digital health products will occur when these new tools and workflows can demonstrate capabilities to enhance patient education, engagement, compliance, and personal agency working in service to scalable, reproducible, and sustainable Shared Decision Making. A new AI-enhanced technology solution platform will require a knowledge management system that facilitates patient-specific data representation in a machine-readable format/architecture for on-demand analysis, reasoning, inferencing, and querying. Ontology-governed personal clinical knowledge graphs allow for this functionality.

b. What changes would enable timely access to high quality CMS and provider generated data on patients?

- PC-5b Response:
    - A new and novel, advanced, patient-level, health information representation platform architecture: Ontology Governed Personal Clinical Knowledge Graphs. These architectural platforms are built to ingest previously siloed, fragment, multi-modal, patient-level, structured and unstructured data/information. The patient's health information is represented according to defined semantic guidelines and structured for advanced analysis by machine deep learning algorithms. The patient's personal health data/information is treated as "Code" and the advanced representation architecture functions as a new patient-level "Operating System".

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

- PC-6 Response:
    - Frictionless and intuitive user interface (UI).
    - Clear and easy to follow user instructions and instructional videos.
    - User-directed, natural language text or voice-activated queries for on-demand health information retrieval.
    - Live, on-demand customer support

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

- PC-7 Response:
    - Incentivize and/or mandate participation in real-world data collection as a condition of CMS reimbursement for new digital health technologies. This could include reporting patient clinical outcomes, treatment/medication adherence, impact on downstream resource utilization (e.g., hospitalizations, ED visits, etc.), clinical end points goals (BP, weight, HbA1c, lipid management, etc.), readmission rates, patient-reported outcomes, shared decision-making metrics, device usage metrics, usability, satisfaction, and quality of life impacts, etc.

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

- PC-8 Response:
  - While a substantial volume of patient health data is technically accessible to patients and caregivers, it remains fragmented across providers, payers, and legacy systems. Information from multiple providers is rarely considered holistically—each provider tends to focus on the data they directly collect or observe, with limited integration of prior records or external inputs. This creates a fractured clinical picture. As a result, the patient—who typically lacks medical training—is often placed in the untenable position of reconciling conflicting information, filling in gaps and making critical decisions based on incomplete or inconsistent inputs. For instance, a cardiologist may recommend a treatment plan without full visibility into recent neurological evaluations or medication changes made by a primary care physician. Each provider sees only a slice of the patient's health story, and the patient is left to assemble the full narrative. To move beyond this fragmented state, health data must be transformed from isolated "data points" into "integrated knowledge". This requires a shift toward modern health information representation platforms—specifically, **ontology-governed personal clinical knowledge graphs**. These systems organize multimodal data (clinical notes, labs, medications, imaging, patient-reported outcomes, etc.) into structured, contextual formats that can be analyzed holistically.

    For example, consider a Medicare beneficiary managing diabetes, hypertension, and mild cognitive impairment. Today, her glucose logs are stored in a mobile app, her medication history sits in her pharmacy record, and her recent neurologist notes exist only in scanned PDFs. No provider sees the full picture, and the patient must navigate her care with limited insight. With a unified knowledge graph, however, this information could be consolidated, semantically aligned, and queried using AI. A prompt like *"What treatment adjustments have improved A1C for*

patients like me with similar comorbidities?" would yield personalized, evidence-based insights—rather than placing the burden on the patient to interpret scattered charts and summaries. In short, data must not only be **available**—it must be **integrated, contextualized, and actionable**. Only then can we move from passive data collection to patient empowerment, provider coordination, and meaningful shared decision-making. CMS is uniquely positioned to catalyze this transformation by promoting interoperable, AI-ready data architectures that respect patient ownership while enabling system-wide intelligence.

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

- PC-8a Response:
    - All the various multi-modal raw data sources listed (and more), when managed in aggregate with the latest frontier data/knowledge representation architectures, deep machine learning algorithms and advanced fine-tuned proprietary LLMs, can be rendered extremely valuable. In the context of the patient, "hard to access personal health data", is a dynamic that UNIVERSALLY AND CATEGORICALLY MUST BE MITIGATED TO ZERO. Patients own their personal health and health-related data. Providers, Payers and commercial EHR's do not.

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

- PC-8b Response:
    - All data that affects directly or indirectly, an individual's health and well-being, should be viewed as valuable. See above.

c. What specific opportunities and challenges exist to improve accessibility, interoperability, and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

- PC-8c Response:
    - Improving accessibility, interoperability, and integration of clinical data across systems requires a foundational shift in how we structure and interpret health information. Much of today's health data exists in structurally interoperable formats—such as standardized HL7 messages or FHIR resources—but lacks the **semantic coherence** necessary for systems to truly understand and reason over it. This leads to major challenges when multiple systems attempt to integrate data for meaningful clinical use. The core issue is not just how data is exchanged, but how it is interpreted. Different providers may use different terminology, omit contextual metadata, or represent the same condition in incompatible formats. Consequently, integrating data from multiple sources often results in duplication, ambiguity, or even contradiction—making it difficult to support advanced clinical decision-making, research, or patient self-management.

      What is needed is a semantic layer that unifies these fragmented inputs and translates raw data into a coherent, machine-readable model. **Ontology-governed personal clinical knowledge graphs** offer such a structure. By organizing patient data as interconnected concepts and relationships—rather than flat lists or free-text notes—these systems enable real-time reasoning, cross-domain queries, and intelligent summarization.

      For example, in a value-based care setting, a patient's medication adherence data (from a pharmacy), glucose readings (from a device), and recent discharge notes (from a hospital EHR) might all be available—but siloed. A **knowledge graph** can harmonize these data streams, surface interactions (e.g., medication changes impacting glucose control), and answer context-

rich questions like: *"Has this patient's cognitive function declined in conjunction with a new medication regime?"* This is the difference between data being **stored** and data being **used**. The opportunity lies in elevating structured health records into **operational knowledge systems** that support **timely, context-aware, and personalized insights** for all stakeholders—patients, providers, researchers, and policymakers.

To realize this opportunity, CMS and partners must:

- Incentivize adoption of semantic data representation standards and architectures.
- Align certification requirements with semantic interoperability—not just transport protocols.
- Support APIs and platforms that expose longitudinal patient models rather than static records.
- Support the development of provider-centric applications that are based on fully integrated knowledge graphs that incorporate the breadth of the patient's clinical experience and eliminate clinical bias from that provider's direct experience.
- Support the development of patient-centric applications to provide an actionable framework for decision making that is based on wholistic information that is presented in a way that enables the patient to make effective decisions in a fragmented clinical environment
- Prioritize open ontologies and modular knowledge graph frameworks that can evolve with clinical evidence.

Ultimately, this is not just an IT transformation—it is a CARE TRANSFORMATION. Enabling meaningful integration of clinical data will unlock insights, reduce duplication, identify patterns, and empower all parties to make better, faster decisions at scale.

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

- PC-12 Response:
  - Referencing the Philosophies, POV's and "Tech Stack" overview provided above (PC-8 Response) and scaling this "Opportunity" across CMS populations, multiple high value use cases emerge:
    - Substantial enhancement of patient-level **understanding, education, engagement, compliance, personal agency**
    - A legitimate, objective, scalable, reproducible, and sustainable roadmap **to Shared Decision Making**
    - A "real world", "information-driven" mechanism to identify and characterize root causes of **"Needless Care Variation"**
    - Given the accepted presence of "Needless Care Variation" and therefore the presence of "Low Value Care", this new knowledge management system will have the computational power and granularity to identify and stratify across the spectrum from real-world "Low Value Care" to "Best-in-Class Care" for any given and highly specific medical condition occurring in highly similar patient cohorts. If it is agreed that "Low Value Care" exists and is identifiable then "Best-in-Class Care" must also exist and be identifiable under specific constraints. If this is true, then real-world **"Best-in-Class Care" pathways, recommendations, guidance could be "Democratized at Scale".**
    - A mechanism to answer universal, real-world patient questions/desires: "**What is the best treatment advise and treatment journey for people like me who have the**

same medical condition that I have?" and, "What works for who (why, when, how) and what are the tradeoffs?"

- A "real-world", "information-driven" **population health management and value-based care knowledge management platform.**
- Improved **clinical outcomes**
- Improved **patient safety**
- Improved **operational efficiency**
- Improved mitigation of **healthcare inequities and disparities**
- ***THE EMERGENCE OF AN AI-POWERED SMART EHR PLATFORM THAT ENABLES SCALABLE, REPRODUCIBLE AND SUSTAINABLE VALUE CREATION FOR PATIENTS, PROVIDERS AND PAYERS.***

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

- PR-1 Response:
  - Build enduring TRUST (performance, intent, integrity) in specific digital health products and platforms. This trust is achieved, supported, and maintained on several levels:
    - Quality, veracity, and fidelity of the foundational, patient-specific data on which the product is built
    - Effective and efficient access and use of the patient-level data (UX/UI functionality)
    - A general understanding of the foundational data representation architecture (ontology-governed personal clinical knowledge graphs)
    - A general understanding of the machine learning models, reasoning engines and LLMs that represent the backend technology stack

- Functionality that facilitates, verifies, and memorializes shared decision-making efforts
- Functionality that delivers on-demand data operationalization capabilities
- Functionality that identifies and promotes real-world "high-value care" therefore mitigating "low-value care", needless care variation, disparities, inequities, and biases
- Functionality that facilitates precision medicine derived from advanced knowledge management systems

a. What are the current obstacles?

- PR-1a Response:
  - On-demand data operationalization is unachievable with current legacy EHR data representation architectures

b. What information should providers share with patients when using digital products in the provision of their care?

- PR-1b Response:
  - The patient should have access to and utilization of the same digital health products and therefore the same information as the provider given that the patient OWNS their health data.

c. What responsibilities do providers have when recommending use of a digital product by a patient?

- PR-1c Response:
  - see PR-1 TRUST (performance, intent, integrity) in the digital health product platform

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

- PR-2 Response:
    - Current commercial EHR platforms have inadequate data/knowledge representation architecture and data operationalization capabilities subsequently preventing time-sensitive, context-aware, point-of care, and point-of task, information dominance and decision advantage that would otherwise support physician workflows. Ontology-governed personal clinical knowledge graphs can achieve what current EHR platforms cannot.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)?

- PR-3 Response:
    - Building a disruptive knowledge management system grounded in ontology-governed, multimodal personal clinical knowledge graphs (PCKGs) represent a transformative shift away from conventional document- or encounter-based EHRs toward semantically rich, context-aware, patient-centered digital ecosystems. At the foundational layer of the technology stack, these knowledge graphs offer structural, semantic, contextual integration, and logical interoperability of disparate (multimodal) healthcare data sources, enabling advanced reasoning, personalization, interoperability, and automation. This new proposed knowledge management system enables AI-supported querying, inferencing, and contextualization of data in a way that traditional relational EHRs cannot. It also enables holistic, longitudinal patient modeling and supports advanced analytics and frontier AI capabilities further advancing patient care quality. A reasoning layer

leverages ontologies to enrich, validate, and infer new knowledge from raw clinical events. Each patient's knowledge graph is individualized, evolves over time and supports:

- Precision medicine: integrates genomics, lifestyle, and response data
- Behavior modeling: incorporates patient adherence, sentiment (via NLP), goals
- Risk models: subgraphs for polypharmacy, frailty, or readmission risk

These proposed new knowledge management systems facilitate movement beyond encounter-based views (current EHR platforms) to living, evolving, and continuously updated health models**.** By embedding ontology-governed, multimodal personal clinical knowledge graphs at the foundational layer of a new and novel knowledge management system, you create a new AI-powered "Smart EHR"—one that is a living, learning, semantically aware health intelligence system that is:

- Patient-centered, patient, provider, payer accessible
- Extensible and interoperable across multimodal data sources
- AI-enabled, machine readable, LLM queryable
- Clinically, operationally, financially valuable
- Built for next-gen value-based care, research, and precision medicine
- Structured for enhanced personal agency and shared decision making
- Structured to identify and mitigate low-value care, disparities, inequities

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

- PR-4 Response:
    - The development of a Patient-Centric, AI-Enhanced, Data-Driven, Platform-Based Healthcare Ecosystem supported by a novel foundational Next-Gem "Smart EHR" Knowledge Management System.

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

- PR-7 Response: see PR-3 Response
    - PR-3 Response: Building a disruptive knowledge management system grounded in ontology-governed, multimodal personal clinical knowledge graphs (PCKGs) represents a transformative shift away from conventional document- or encounter-based EHRs toward semantically rich, context-aware, patient-centered digital ecosystems. At the foundational layer of the technology stack, these knowledge graphs offer structural, semantic, contextual integration, and logical interoperability of disparate (multimodal) healthcare data sources, enabling advanced reasoning, personalization, interoperability, and automation. This new proposed knowledge management system enables AI-supported querying, inferencing, and contextualization of data in a way that traditional relational EHRs cannot. It also enables holistic, longitudinal patient modeling and supports advanced analytics and frontier AI capabilities further advancing patient care quality. A

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PA-5. What are ways payers can help with simplifying clinical quality data responsibilities of providers?

a. **How interested are payers and providers in EHR technology advances that enable bulk extraction of clinical quality data from EHRs to payers to allow them to do the calculations instead of the provider-side technology?**

b. **In what ways can the interoperability and quality reporting responsibilities of providers to both CMS and other payers be consolidated so investments can be dually purposed? Are there technologies payers might leverage that would support access to real time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?**

- PA-5a,b Response:
    - In a "Perfect World" and in the context of "First Principles", each of the three major stakeholders occupying the center of the healthcare ecosystem (patients, providers, and payers) would have access to the same knowledge management system. This patient-centered, ontology-governed personal clinical knowledge graph (PCKG) data architecture would function as the single source of truth for patient, provider, and payer. It would be a living, constantly evolving, and persistently updated digital, multimodal data representation of the patient that offers multiple unique value propositions for each stakeholder. When multimodal, patient-level, health and health-related data is represented as proposed, the patient's complete longitudinal healthcare journey is modeled in an interoperable (logical), structured digital format that facilitates:
        - advanced AI-supported (machine learning) analysis
        - sophisticated reasoning engines enabling inference
        - use of frontier LLMs for on-demand personal health record querying

Now and only now, can the questions (PA-5a,b) be answered in an affirmative, time-sensitive, context-aware, stakeholder specific, scalable, reproducible, sustainable, transparent, explainable, and uniform fashion. We are quickly approaching the "AI Agent Economy" which will facilitate the embedding of human-level decision-making into autonomous software. These coming AI agents will contain a semantic capability layer within the technology stack that will allow them to interpret context, make decisions, and perform tasks with minimal human supervision. In summary, these AI multi-specialty agents will be able to listen, reason, decide and act! Implicit in an upcoming AI Agent Era, is the presence of a necessary and sufficient "knowledge management system" over which these agents can fully exploit their stated and desired capabilities. The current EHR data/information architecture will not support these goals. Issues of data governance, scope of data access, digital security, privacy compliance, digital-user monitoring," read only" status, "read and write back to" status, platform ecosystem architecture, as well as other "systemic matters" require further thoughtful and vigilant consideration.

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

- TD-1 Response:
    - CMS must support, facilitate, incentivize, and even mandate, data-driven, technology-enhanced, objective and scaled Shared Decision Making (SDM). If CMS beneficiaries are truly to be placed at the center of the healthcare ecosystem, they must be empowered (educated, engaged, given agency) at the beginning of their personal healthcare journey(s). Treatment decisions must be heavily influenced and informed by quality-of-life and outcome goals important to the patient. Patient-level, SDM is often the topic of well-intended discussion however, there is NO objective infrastructure platform that supports and facilitates scalable, reproducible, and sustainable SDM.

In addition, emphasis must be directed towards objective, data-driven, technology-enhanced mitigation of:

- Needless Care Variation
- Delivery of Low Value Care
- Healthcare Inequities and Disparities.

These systemic and pervasive healthcare dynamics have risen to a societal level of importance. These dynamics contribute a substantial portion of wasteful healthcare spending in the U.S., (currently estimated at $1-1.25 Trillion/year-U.S.). Solutions to these widespread issues will require a new and novel "knowledge management system" capable of representing, managing, and operationalizing multi-modal, personal, patient-level, health and health-related data. "Information Dominance" and "Decision Advantage" should be the expected, desired, and valuable "Output Exhaust" from these new "knowledge management systems". Solving large and extremely important (societal) problems, the results of which, facilitate objective, observable, and tangible life-altering health, and wellness benefits, is the stimulation (for developer interest in building digital health products for Medicare beneficiaries and caregivers) CMS is seeking.

TD-2. Regarding CMS Data, to stimulate developer interest--

a. What additional data would be most valuable if made available through CMS APIs?

- TD-2a Response:
  - All available patient-level data is valuable for the population of new and novel Knowledge Management Systems:
    - Ontology-governed personal clinical knowledge graphs
    - Machine deep learning models

- AI-enhanced reasoning engines
- Sophisticated fine-tuned, prompt augmented, secure LLMs

c. **What obstacles prevent accessing these data sources today?**

- TD-2c Response:
  - Status quo mindset
  - Bureaucratic inertia
  - Subtle and pervasive data blocking practices
  - Legacy IT infrastructure

d. **What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?**

- TD-2d Response:
  - All those APIs required to serve the best interest of CMS's beneficiary population base. The patient must be the central focus of all decisions and initiatives. Both patients and providers benefit from true operationalized, interoperable (structural and logical), patient-level health and health-related data. Patients and providers benefit from time-relevant, context-aware, "information dominance" and "decision advantage"- both achievable with new and novel "knowledge management systems". The current EHI data architecture is incapable of performing against these constraints.

    The ASTP will need to facilitate and support a substantially new data architecture platform into which (scaled) patient-level EHI can be represented, visualized, and analyzed if their vision, mission, and strategic goals are to be realized. The current problems hampering the stated vision, mission and strategic goals are vastly deeper and more complex than choosing additional API's. From a "First Principles" perspective, a new data architecture platform and a new knowledge

management system will both be required to unleash AI-based innovation that supports patients, providers, and payers.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

- TD-13 Response:
    - Access to the entirety of a patients EHI would provide the opportunity to reconfigure the representation of the patient's health and health-related data. If a patient's EHI is fully represented by ontology-governed personal clinical knowledge graphs (KG's), then the data can be considered a structured, semantically rich form of "code", unlocking several powerful capabilities. Knowledge graphs transform raw clinical data (labs, provider notes, test results, diagnoses) into standardized, interconnected entities and multidimensional relationships. Machine learning models/algorithms can perform predictive functions, time-sensitive reasoning, and learn topological data patterns and semantic relationships. Ontological governance allows symbolic AI reasoning and supports explainable decision making. LLMs fine-tuned or prompt-augmented on available medical ontologies can provide clinical summarization and natural language reasoning grounded in ontological structure. Ontology-governed knowledge graphs enforce semantic (logical) data interoperability and facilitate advanced data operationalization (AI machine-learning analysis, reasoning, inference, and querying capabilities) ultimately providing "information dominance" and "decision advantage"-all in service to the patient.

a. **What are the primary obstacles to this?**

- TD-13a Response:
  - Access to patient-level EHI data at sufficient scale to build out required paradigm-changing data representation models/architectures and new knowledge management systems.

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients?

- VB-1 Response:
  - Digital health management must fundamentally start with optimal data integration, interoperability, representation, visualization, and operationalization. While databases (current EHR's) impose structural constraints, they do not define logical rules or knowledge. Meaning is not inherent in a (EHR) database — it must be provided by ontologies, reasoning engines, and knowledge management systems. Consider the difference between a graph database, a knowledge base, and a knowledge management system (KMS):
    - A graph database stores relationships explicitly but does not infer new ones.
    - A knowledge base contains structured knowledge with defined ontologies (semantics) and reasoning rules.
    - A knowledge management system (KMS) integrates various sources, facilitates reasoning, and supports decision-making by capturing and processing knowledge dynamically.

    In a knowledge management system (KMS):
    - The system integrates multiple knowledge sources.
    - It processes updates dynamically, ensuring that derived insights are maintained over time.

- Users interact with the system through queries, reasoning, and decision-support tools. APM's must be able to operationalize patient-level and patient cohort-level data in a time-dependent, context-aware fashion to make better, faster, point-of-care and forward-looking decisions at scale. Current commercial EHR data architectures do not support these necessary capabilities. Optimal data operationalization fundamentally requires understanding structural vs. logical interoperability, and distinguishing between databases, knowledge bases, and knowledge management systems This understanding is crucial for designing systems that are efficient and effective at driving information dominance and decision advantage. To truly achieve health-data interoperability and knowledge-driven systems, there is an immediate need for semantic layers (ontologies), reasoning engines, and new knowledge frameworks. When these new capabilities are firmly in place, optimal data operationalization can be an expected outcome-----BETTER, FASTER, SMARTER DECISIONS AT SCALE IS THE ULTIMATE INCENTIVE.

a. What are the current obstacles preventing broader digital product adoption for patients in ACOs?

- VB-1a Response:
  - The "inflexibility" and "isolation" of the current EHR systems

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

- VB-2 Response:
    - The key themes and technologies referenced can only be optimally leveraged and integrated into APM requirements when the foundational patient-level data, from which all decisions are derived, is functionally operationalized in a time-sensitive, context-aware, point-of-decision fashion. Current commercial EHR data architectures do not support functional data operationalization. To solve these problems an "ontology" is needed. The ontology provides the semantic (architectural) blueprint for the development of a modern-day, AI-ready, data representation platform---the knowledge graph. When patient-level data is structured in this format, AI-augmented machine learning models and advanced fine-tuned and prompt-augmented LLM queries can be developed and leveraged.

VB-3. What are essential health IT capabilities for value-based care arrangements?

- VB-3 Response: See VB-1 and VB-2

VB-4. What are the essential data types needed for successful participation in value-based care arrangements?

- VB-4 Response:
    - All available patient-level clinical data, claims and administrative data, patient-reported outcome data, social determinates of health data, care coordination and longitudinal data, quality measures

and benchmarking data, cost and financial data, real-time and remote monitoring biometric data. For optimal VBC platform performance, there must be structural and logical data interoperability as well as sophisticated (AI-enabled) data analysis that supports risk stratification, care gap assessment, care coordination, and outcome performance and attribution.

A REPEATING THEME: ON-DEMAND INFORMATION DOMINANCE AND DECISION ADVANTAGE AT SCALE!

For more information, please contact:

David Theodoro, MD, MBA
CEO, Recorded Health Inc.
info@recordedhealth.com
dtheodoro@me.com